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ChE 312 Example Exam 1: Closed-book section

30 points, 4 problems, 2 pages. [Our blank page if you need it.]

Write your name on each page, representing your honor commitment.

Be specific about units, watch significant digits, show/explain your calcs. **NEVER ERASE!**

Prob. 1 (10 pts). Define "species volatility" and "relative volatility" for vapor-liquid equilibrium, both in words and with mole-fraction variables. Are these definitions valid for non-ideal gases and liquids?

Volatility (or distribution coefficient K_i) is y_i/x_i . For a chemical species, its volatility is how high its mole fraction is in the gas phase relative to the liquid phase. If it is highly volatile, $y_i \gg x_i$, and if its volatility is low, $y_i \ll x_i$.

Relative volatility is $\alpha_{12} = (y_1/x_1)/(y_2/x_2)$. It compares volatility of species 1 relative to some reference species 2, usually choosing the most-volatile species as #1.

Yes, these definitions valid for non-ideal gases and liquids because they make no assumptions of ideality.

[Remark: If a phase is nonideal, the same compound may not always be more volatile, such as happens for an azeotrope.]

Prob. 2 (5 pts). To make a distillation column work, what is the purpose of a condenser at the top?

The condenser generates liquid to be refluxed back into the column, providing liquid for vapor-liquid countercontacting.

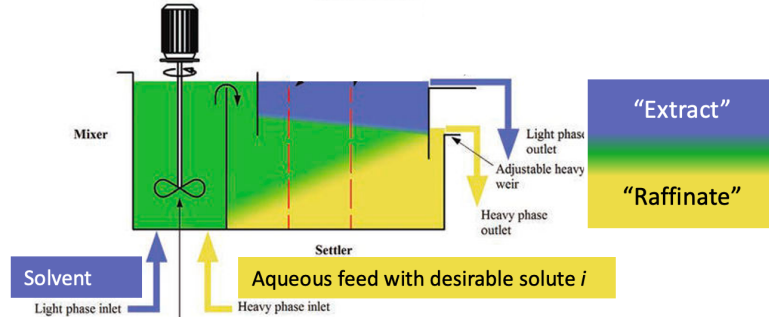
[Liquid or saturated vapor may be generated by the condenser.]

Prob. 3 (5 pts). To make a distillation column work, what is the purpose of a reboiler at the bottom?

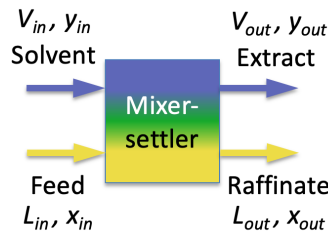
The reboiler generates vapor to feed into the bottom of the column, providing vapor for vapor-liquid countercontacting.

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Prob. 4 (10 pts). A "mixer-settler" for continuous-flow liquid-liquid extraction (see the figure) is just like a flash drum but with two feeds that get mixed together. Two liquids, a lower-density organic solvent ("light phase") and a higher-density aqueous liquid ("heavy phase") are fed and mixed. As the two immiscible phases re-separate, a compound i that is soluble in both is transferred to the solvent "extract" phase where it has a higher solubility. The solute-depleted aqueous phase is called the "raffinate." Equilibrium mole fractions for the solute are related by $\gamma_{i,\text{light}}x_{i,\text{light}} = \gamma_{i,\text{heavy}}x_{i,\text{heavy}}$.



Sketch a control volume around the whole mixer-settler; define suitable flow and mole-fraction variables; and write an overall balance and a solute-species balance.



You can choose any variable names you want, so long as you have flow and mole-fraction variables and can indicate which are going in and which are going out.

Then

$$\begin{aligned} L_{in} + V_{in} &= L_{out} + V_{out} \\ L_{in}x_{in} + V_{in}y_{in} &= L_{out}x_{out} + V_{out}y_{out} \end{aligned}$$

or

$$L_{in}x_{i,\text{heavy},in} + V_{in}x_{i,\text{light},in} = L_{out}x_{i,\text{heavy},out} + V_{out}x_{i,\text{light},out}$$

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ChE 312-001 Exam 1, Spring 2023: Open-book section

70 pts, 3 problems (5, 6, 7); 3 pp with graph; extra blank page.

Write your name on each page, representing your honor commitment.

Be specific about units, watch significant digits, show/explain your calcs. **NEVER ERASE!**

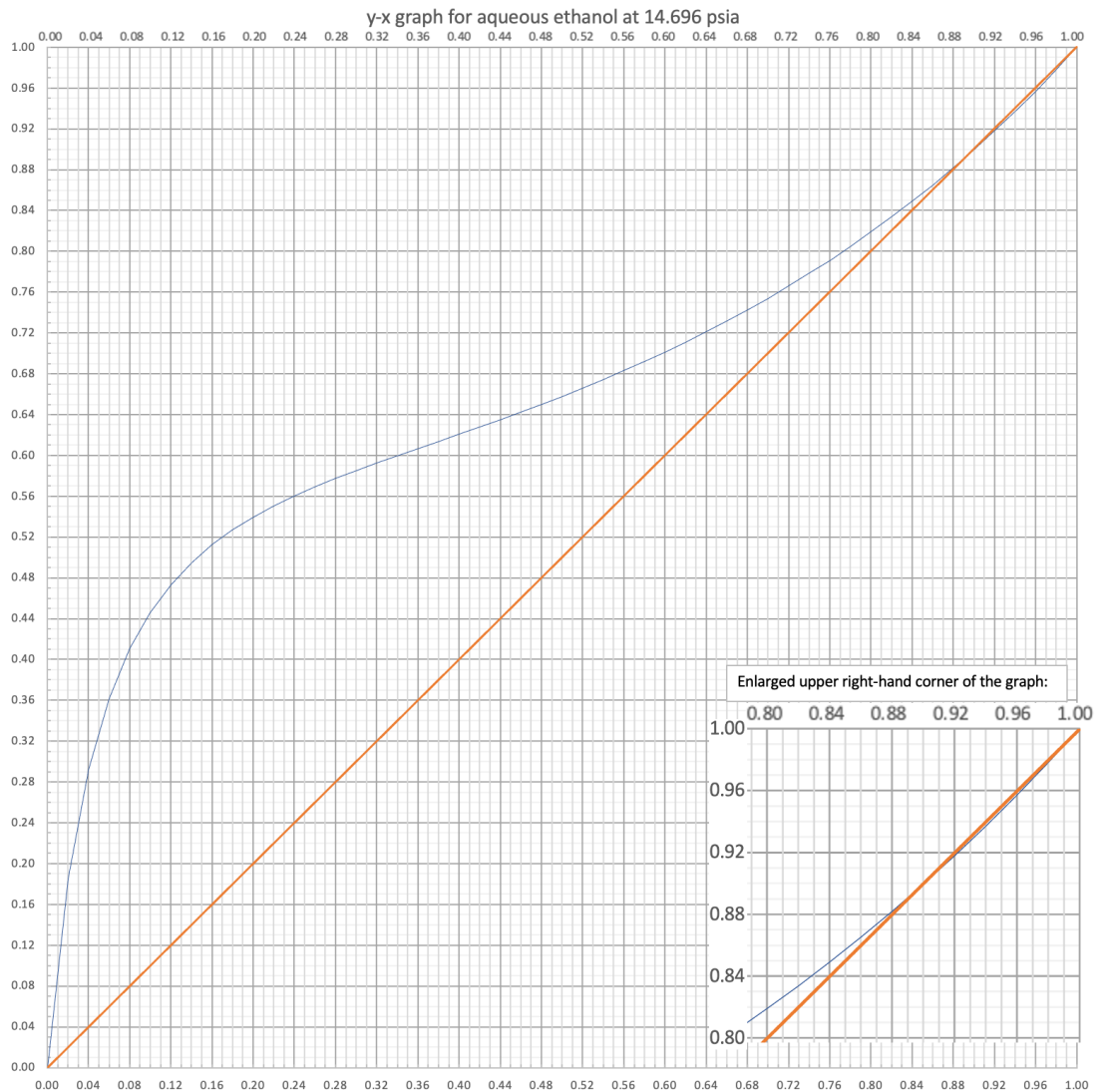
Problem 5 (40 pts).

Purifying bio-based fuels and chemicals is often complicated by their oxygen content. An example is distillation of ethanol-water mixtures, which are nonideal enough to have a minimum-boiling azeotrope at $x_{\text{ethanol}}=0.90$. That's okay; we can distill from feeds having $z_{\text{ethanol}} < 0.90$.

Fermentation ethanol is up to about 12-14 mol% in water. Design a column for 100 mol/h of saturated-liquid feed of $z_{\text{ethanol}}=0.120$ to make distillate having $x_{\text{ethanol}}=0.82$. There is no specification on the bottoms concentration, but 95% of the feed ethanol should go into the distillate. Equilibrium mole fractions are provided on the attached graph. Use a total condenser and a partial reboiler.

Determine the optimum reflux ratio based on $R=1.5 R_{\text{min}}$; the number of stages; number of trays; and optimal feed stage location. Fill out the table of mole fractions using McCabe-Thiele analysis.

Stage	x	y
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
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15		
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First, sketch what you know.

To get D , W , x_W , use "95% of the feed ethanol should go into the distillate."

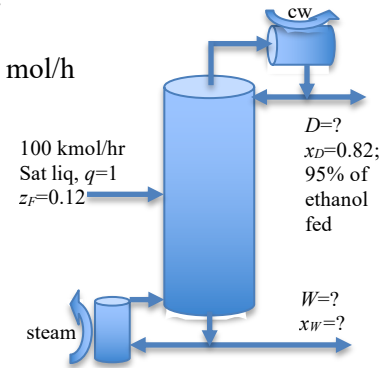
The feed ethanol is $z_F F = (0.12)(100) = 12$ mol/h ethanol.

95% into D implies $(0.95)(12) = 11.4$ mol ethanol/h in D , so $D = 11.4/0.82 = 13.9$ mol/h

From $F = D + W$, then $W = 100 - 13.9 = 86.1$ mol/h

And from $z_F F = x_D D + x_W W = (0.12)(100) = (0.82)(13.9) + x_W (86.1)$

then $x_W = 0.0070$



The enriching-section mass balance (OL) has a tangent pinch such that the limiting OL goes through $(0.82, 0.82)$ and $(0, 0.347)$.

Its slope then is $\frac{R_{min}}{R_{min}+1} = \frac{0.82-0.347}{0.82-0} = 0.577$

Solve to get $R_{min} = 0.577R_{min} + 0.577$, so $R_{min} = 1.363$

Then $R = 1.5R_{min} = 2.044$ and $\frac{R}{R+1} = 0.672$

The ES OL is $y_{n+1} = \frac{R}{R+1}x_n + \frac{x_D}{R+1} = 0.672x_n + 0.27$, so it goes through $(0.82, 0.82)$ and $(0, 0.27)$.

The feed-quality line is $y = \frac{q}{q-1}x - \frac{z_F}{q-1}$, so it is a vertical line at $x = 0.12$

The lines then intersect at $y = 0.672(0.12) + 0.27 = 0.35$; that is, at $(0.12, 0.35)$.

Also, the stripping-section OL goes through $(0.12, 0.35)$ and $(0.0067, 0.0067)$, so $y_{n+1} = 3.03x_n - 0.014$

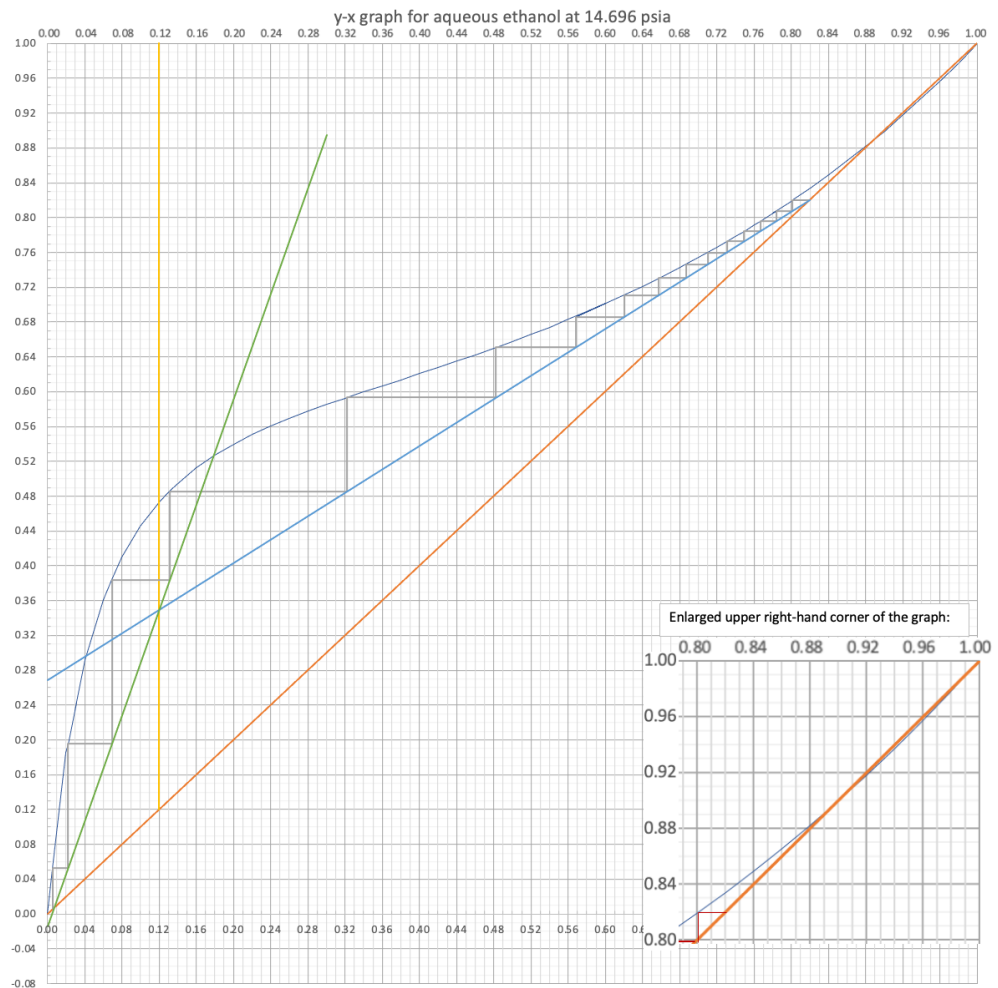
Starting at $y_1 = 0.82$, find the equilibrium x_1 from the graph.

Then find y_1 from the graph or the ES OL equation, repeating until the next x_{eq} is less than 0.12 (the feed stage).

Shift to use the stripping-section OL equation or graph until the next x_{eq} is less than 0.0067.

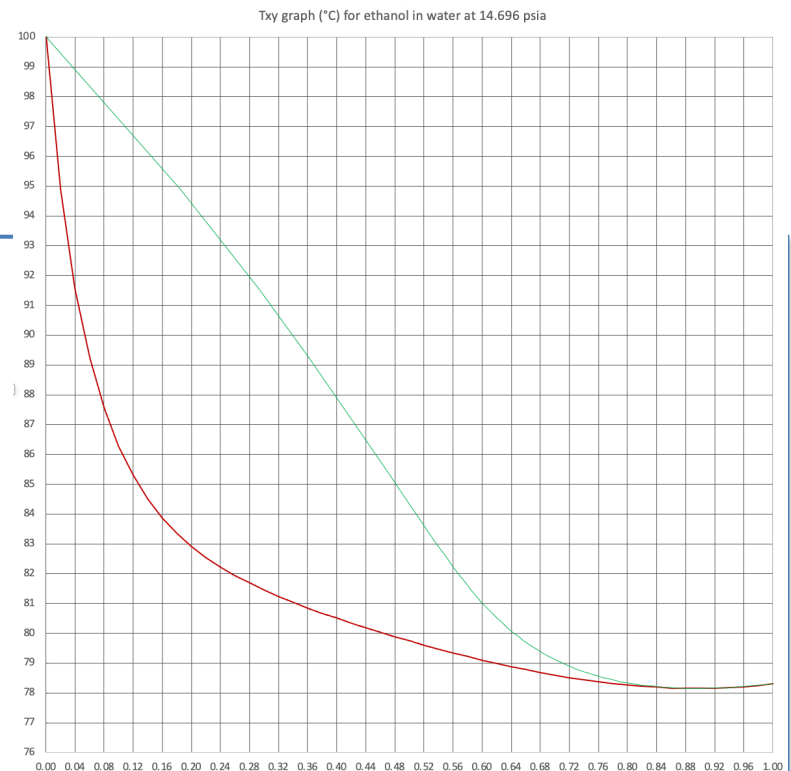
The result is 16 stages (15 trays plus a reboiler) with optimal feed stage 13.

Stage	x	y
1	0.801	0.820
2	0.784	0.808
3	0.767	0.796
4	0.749	0.784
5	0.731	0.773
6	0.710	0.760
7	0.687	0.746
8	0.658	0.730
9	0.620	0.711
10	0.568	0.686
11	0.482	0.651
12	0.322	0.593
13	0.131	0.485
14	0.069	0.384
15	0.022	0.196
16	0.006	0.053



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Problem 6 (20 pts). Using the Txy equilibrium diagram at right, determine the approximate temperature at the reboiler, the feed tray, and the top tray. Explain why the temperature changes as it does from the reboiler to the feed tray to the top tray.



Top: (0.802,0.84) 78.3°C
 Feed: (0.098,0.440) 86.5°C
 Reboiler: (0.0001,0.015) 99.9°C

The equilibrium temperature changes as the more-volatile compound becomes higher in concentration (as you go up). The top is coolest because it is high in volatile ethanol, whose pure boiling point is close to 78.3 despite the azeotrope. The reboiler is nearly pure water, whose boiling point is 100°C.

Problem 7 (10 pts). The idea of a K_i value in binary or multi-component equilibrium is to compare mole fractions for chemical species i in two equilibrium phases, whether vapor-liquid, liquid-liquid, liquid-solid, vapor-solid, or solid-solid. In equilibrium between an ideal gas mixture and an ideal liquid solution involving multiple compounds, write an equation for K_i for chemical species i in terms of phase mole fractions, pressure, and temperature.

For VLE, $K_i = y_i/x_i$, the same expression as for volatility.

For an ideal gas mixture and an ideal liquid solution, Raoult's Law is valid: $y_i P = x_i P_{vap,i}(T)$,

so $K_i = y_i/x_i = P_{vap,i}(T)/P$.

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